



RADIOLOGY SUBJECT WISE TEST DATED ON 31st MAY 2026

QUESTIONS

- Q1. A 65-year-old hypertensive patient on irregular medication presents with sudden severe headache and BP of 220/110 mmHg. CT head (non-contrast) done is shown below. The MOST likely diagnosis is:



- a) Intraventricular haemorrhage
- b) Putaminal haemorrhage
- c) Subarachnoid haemorrhage
- d) Pontine haemorrhage

- Q2. A 33-year-old female presents with fever, weight loss, dry cough and elevated serum ACE levels. Chest X-ray is shown below. The MOST likely diagnosis is:



- a) Primary tuberculosis
- b) Sarcoidosis
- c) Silicosis
- d) Lymphoma

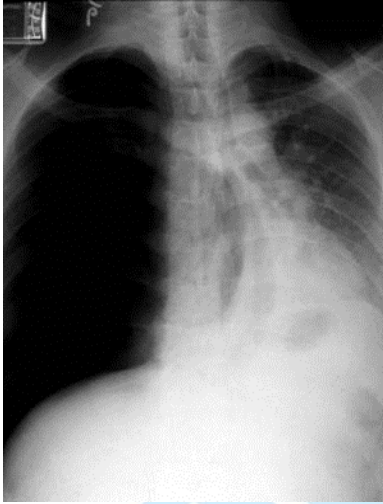
- Q3. A 50-year-old male road-traffic accident victim is brought unconscious to the casualty. CT head shows hyperdense crescent-shaped collection along the inner table of the right parietal bone without crossing the midline, associated with midline shift to the left. The MOST likely diagnosis is:

- a) Extradural haematoma
- b) Chronic subdural haematoma
- c) Acute subdural haematoma
- d) Intracerebral contusion

- Q4. A patient post RTA presents to the casualty with hypotension and tachycardia. He can speak in words BUT cannot complete sentences. CXR done is shown below. The MOST likely diagnosis is:



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- a) Right tension pneumothorax
- b) Left pleural effusion
- c) Right pulmonary collapse
- d) Left diaphragmatic hernia

Q5. Regarding MRI signal characteristics, which of the following statements about T1- weighted (T1W) and T2-weighted (T2W) images is CORRECT?

- a) CSF appears bright on T1W and dark on T2W
- b) Fat appears bright on T1W and bright on T2W
- c) Chronic hematoma appears dark on T1W and bright on T2w image
- d) Water appears bright on T1W

Q6. A patient presents with a hard palpable gallbladder, painless jaundice and weight loss (Courvoisier's sign). Ultrasound abdomen reveals a dilated common bile duct and gallbladder with a hypoechoic mass at the head of the pancreas. The MOST appropriate NEXT investigation to delineate the extent of disease is:

- a) ERCP alone
- b) Contrast-enhanced CT abdomen (CECT) with pancreatic protocol
- c) Plain X-ray abdomen
- d) HIDA scan

Q7. A patient who had a road traffic accident has petechial haemorrhages on CT head but is unconscious with a GCS of 4. No mass lesion or midline shift is seen. The MOST likely diagnosis is:

- a) Concussion
- b) Cerebral contusion
- c) Diffuse axonal injury
- d) Subarachnoid haemorrhage

Q8. A patient with prostate cancer undergoes radiotherapy at a dose of 1.8–2 Gy per fraction, 5 fractions per week, over 7–8 weeks (total 70 Gy). This radiotherapy schedule is called:

- a) Hypofractionated radiotherapy
- b) Hyperfractionated radiotherapy



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- c) Conventional (standard) fractionated radiotherapy
d) Stereotactic body radiotherapy (SBRT)
- Q9. A 55-year-old male with known pulmonary tuberculosis on treatment presents with new-onset haemoptysis. Chest X-ray shows a cavity in the right upper lobe containing a rounded opacity with a crescent of air around it. The MOST likely aetiology is:
- a) Aspergilloma (mycetoma)
b) Lung abscess
c) Hydatid cyst
d) Squamous cell carcinoma
- Q10. Gold standard investigation for pre-operative planning of functional endoscopic sinus surgery (FESS) is:
- a) Plain X-ray paranasal sinuses (Waters' view)
b) MRI paranasal sinuses
c) CT paranasal sinuses (coronal plane, bone window)
d) Nasal endoscopy alone
- Q11. A screening mammogram of an asymptomatic 52-year-old woman shows a stellate (spiculated) high-density lesion with architectural distortion in the upper outer quadrant of the left breast, with associated micro-calcifications. According to the ACR BI-RADS classification, this lesion is BEST categorised as:
- a) BI-RADS 2 – Benign
b) BI-RADS 3 – Probably benign
c) BI-RADS 4B – Moderate suspicion
d) BI-RADS 5 – Highly suggestive of malignancy
- Q12. A 60-year-old male heavy smoker presents with haemoptysis, weight loss and hoarseness of voice. CXR shows a hilar mass with post-obstructive collapse of the right upper lobe. CT chest confirms a spiculated right hilar mass invading the mediastinum. The BEST staging investigation for this patient is:
- a) CECT chest abdomen pelvis alone
b) PET-CT (FDG PET-CT) whole body
c) MRI chest
d) Bone scan + CECT chest
- Q13. A 30-year-old male presents with haematuria and right flank pain. KUB X-ray shows a 7 mm radio-opaque calculus in the right ureter at the level of L3. Non-contrast CT KUB (NCCT KUB) is performed. Which is the MOST accurate first-line imaging for diagnosis and characterisation of urolithiasis?
- a) Intravenous urography (IVU)
b) Ultrasound KUB
c) Non-contrast CT KUB
d) MRI urography



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- Q14. A 50-year-old diabetic male presents with painful red swelling of the right foot with black discolouration of the 2nd toe. Plain X-ray of the foot shows cortical destruction and periosteal reaction of the 2nd metatarsal head. The SINGLE BEST investigation to distinguish osteomyelitis from neuropathic (Charcot) joint is:
- a) Plain X-ray foot (repeat)
 - b) Technetium-99m MDP bone scan
 - c) MRI foot with gadolinium contrast
 - d) Gallium-67 scan
- Q15. A 50-year-old male presents to the emergency department following a high-speed motor vehicle accident with hypotension (BP 80/50) and abdominal tenderness. FAST (Focused Assessment with Sonography in Trauma) ultrasound shows free fluid in the hepatorenal space (Morison's pouch). The MOST appropriate next step in this haemodynamically unstable patient is:
- a) CECT abdomen to characterise the injury
 - b) Immediate diagnostic peritoneal lavage (DPL)
 - c) Emergency exploratory laparotomy
 - d) Repeated FAST examination after 30 minutes
- Q16. A 70-year-old male presents with acute onset severe abdominal pain, distension, and absolute constipation. Abdominal X-ray shows a massively dilated loop of large bowel arising from the left iliac fossa, with a characteristic 'coffee bean' / 'bent inner tube' appearance pointing towards the right upper quadrant. The MOST likely diagnosis is:
- a) Caecal volvulus
 - b) Sigmoid volvulus
 - c) Large bowel obstruction due to carcinoma
 - d) Acute colonic pseudo-obstruction (Ogilvie syndrome)
- Q17. A neonate is born at term with bilious vomiting on day 1 of life. Abdominal X-ray done is shown below. The MOST likely diagnosis is:



- a) Pyloric stenosis
- b) Jejunal atresia
- c) Duodenal atresia
- d) Meconium ileus



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- Q18. A 6-week-old male infant presents with non-bilious projectile vomiting after every feed. The vomitus is never bilious. The infant is hungry immediately after vomiting. Ultrasound abdomen shows pyloric muscle thickness of 4.2 mm and pyloric channel length of 17 mm. The MOST likely diagnosis is:
- Duodenal atresia
 - Gastroesophageal reflux
 - Hypertrophic pyloric stenosis
 - Malrotation with volvulus
- Q19. A 9-year-old child presents with acute right iliac fossa pain, vomiting and fever. Ultrasound abdomen shows a non-compressible aperistaltic tubular structure of diameter 9 mm with echogenic periappendiceal fat and a hyperechoic faecolith in the lumen. The MOST likely diagnosis is:
- Mesenteric adenitis
 - Acute appendicitis
 - Meckel's diverticulitis
 - Ovarian torsion
- Q20. A 10-month-old infant presents with intermittent paroxysmal colicky abdominal pain, vomiting and 'redcurrant jelly' stools. Ultrasound abdomen shows a 'target sign' (or 'doughnut sign') in the right upper quadrant. The MOST appropriate NEXT management step is:
- Emergency surgical exploration
 - Air/water enema reduction (pneumatic/hydrostatic reduction) under fluoroscopy or ultrasound guidance
 - MRI Abdomen
 - Technetium-99m Meckel's scan
- Q21. A 2-year-old child presents with a cough, wheeze and sudden onset respiratory distress after playing with small toys. CXR in inspiration appears normal. Expiratory chest X-ray shows hyperinflation of the right lung with mediastinal shift to the LEFT. The MOST likely diagnosis is:
- Right-sided pneumonia
 - Right bronchial foreign body
 - Left tension pneumothorax
 - Right pleural effusion
- Q22. Apple core deformity on barium study in a patient with dysphagia is seen in :
- Carcinoma colon
 - Carcinoma caecum
 - Carcinoma Esophagus
 - Achalasia Cardia
- Q23. The most RADIORESISTANT phase of the cell cycle is:
- G2-M phase
 - S phase
 - G1 phase
 - M phase



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Q24. A 'bat-wing' or 'butterfly' pattern of opacification on CXR is most consistent with?

- a) Pulmonary fibrosis
- b) Cardiogenic pulmonary oedema
- c) Sarcoidosis
- d) Pneumocystis jirovecii pneumonia

Q25. A solitary pulmonary nodule with 'popcorn' calcification is most likely?

- a) Carcinoid tumour
- b) Hamartoma
- c) Carcinoma
- d) Granuloma

Q26. 'Signet ring sign' on CT chest indicates?

- a) Pulmonary emphysema
- b) Bronchiectasis
- c) Pulmonary AVM
- d) Lymphangioliomyomatosis

Q27. 'Double duct sign' on MRCP indicates obstruction at the?

- a) Common hepatic duct
- b) Cystic duct
- c) Ampulla of Vater / pancreatic head
- d) Intrahepatic bile ducts

Q28. CT abdomen shows 'sentinel loop' near the epigastrium in a patient with elevated serum amylase. Significance?

- a) Small bowel obstruction
- b) Localised ileus adjacent to inflamed pancreas
- c) Toxic megacolon
- d) Paralytic ileus from peritonitis

Q29. 'Rugger jersey spine' on plain film is seen in?

- a) Osteoporosis
- b) Renal osteodystrophy
- c) Paget disease
- d) Multiple myeloma

Q30. Which BI-RADS category mandates tissue sampling (biopsy)?

- a) BI-RADS 3
- b) BI-RADS 4
- c) BI-RADS 2
- d) BI-RADS 0

Q31. On DMSA renal scan, cortical defects that persist at 6 months post-UTI indicate?



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- a) Active pyelonephritis
- b) Renal scarring
- c) Renal dysplasia
- d) Normal variant

Q32. Which nuclear medicine scan is BEST for detecting pheochromocytoma?

- a) Tc-99m MIBI
- b) I-131 MIBG scintigraphy
- c) Ga-68 DOTATATE PET
- d) F-18 FDG PET

Q33. Which testicular tumour responds best to radiotherapy?

- a) Teratoma
- b) Seminoma
- c) Choriocarcinoma
- d) Embryonal carcinoma

Q34. Investigation of choice for local staging of oesophageal carcinoma is:

- a) CECT thorax and abdomen
- b) Endoscopic ultrasound (EUS)
- c) PET-CT
- d) HRCT chest

Q35. Which cardiac silhouette shape on CXR is associated with Tetralogy of Fallot?

- a) 'Figure of 8' heart
- b) 'Box-shaped' heart
- c) 'Boot-shaped' heart
- d) 'Egg on side' heart

Q36. A 27-year-old male presents with an incidental finding on spinal X-ray — a vertebral body with a 'jail bar' / corduroy appearance. What is the diagnosis?

- a) Vertebral metastasis
- b) Multiple myeloma
- c) Vertebral haemangioma
- d) Eosinophilic granuloma

Q37. The following is an axial CECT abdomen at the level of the porta hepatis. Identify the structure marked by the arrow:



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- a) Common hepatic artery
- b) Common bile duct
- c) Portal vein
- d) Inferior vena cava

Q38. Identify the correct combination of investigation-disease process?



- a) IVP- Neurogenic Bladder
- b) IVP- Posterior Urethral Valve
- c) IVP – Ureterocele
- d) X-ray - Calculus

Q39. A radiograph is done of an adult male presenting with acute onset abdominal pain. What is the diagnosis?



- a) Lung Abscess
- b) Intestinal Perforation
- c) Splenic Infarct
- d) Pneumothorax

Q40. A Chest X-ray of an adult male presented with respiratory distress. What is the diagnosis?



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- a) Hydropneumothorax
- b) Pleural effusion
- c) Pneumothorax
- d) Consolidation of right lower lobe

Q41. Plain radiograph is done for an alcoholic adult presenting with recurrent pain abdomen and vomiting. What is the cause of the area of calcification in this image?



- a) Calcified gallstones
- b) Calcified mesenteric lymph nodes
- c) Pancreatic calcification
- d) Renal calcification

Q42. The radiograph of an elderly patient coming for evaluation of backache shows the following abnormality. Identify the disease process?



- a) Meningioma
- b) Struge weber syndrome
- c) Plasmacytoma
- d) Neurofibroma

Q43. Imaging done for the evaluation of the lump in the upper abdomen shows the following abnormality. Identify the correct combination of investigation and disease process?



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- a) MRI- Liver Calcification
- b) CT- Hepatic Calcification
- c) CT- Hydatid cyst
- d) MRI- Hydatid cyst

Q44. USG can diagnose all of the following antenatally Except?

- a) Anencephaly
- b) Open Neural Tube Defect
- c) Placenta praevia
- d) Down's Syndrome

Q45. A young male presents with fever, cough with expectoration and shortness of breath. A CXR is done and shows consolidation. Identify the location of the abnormality on the given CXR?



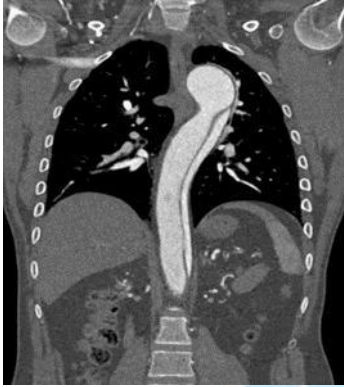
- a) Right Upper Lobe
- b) Right middle lobe, medial segment
- c) Right Middle lobe, Lateral Segment
- d) Right Lower Lobe

Q46. A 50 year old hypertensive patient was brought to the emergency block with acute chest pain. He had tearing sensation in his chest and his ECG and Trop T were negative. A postero-anterior chest radiograph was done to rule out pneumothorax. Patient was taken up for a triple rule out CT angiography and the study revealed the findings as shown in figure below. What is the probable diagnosis?



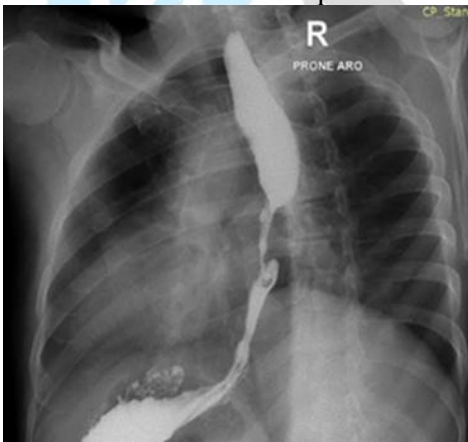
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- a) Pulmonary Thromboembolism
- b) Aortic Dissection
- c) Aortic Stenosis
- d) Aortic rupture

Q47. A 68-year-old male patient came with complaints of dysphagia since past 6 months, which is more for solid as compared to liquid diet. He also complains of significant unintentional weight loss. His barium swallow examination was performed under fluoroscopy and spot images have been shown. Likely diagnosis is?



- a) Benign esophageal stricture
- b) Dysphagia lusoria
- c) Ca esophagus
- d) Mediastinal lymphadenopathy compressing the esophagus.

Q48. LINAC produces?

- a) X-rays
- b) Electrons
- c) Photon & Particulate beam
- d) Proton beam

Q49. Identify the location of the stent ?



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- a) Rt Ureter
- b) Rt Pelvi calyceal system
- c) CBD
- d) Gall bladder

Q50. Gold standard investigation for diagnosis of uterine anomalies is ?

- a) Hysterosalpingography
- b) Laparoscopy
- c) MRI Pelvis
- d) 3D transvaginal sonography

ANSWERS

1	b
2	b
3	c
4	a
5	b
6	b
7	c
8	c
9	a
10	c

11	d
12	b
13	c
14	c
15	c
16	b
17	c
18	c
19	b
20	b

21	b
22	c
23	b
24	b
25	b
26	b
27	c
28	b
29	b
30	b

31	b
32	b
33	b
34	b
35	c
36	c
37	c
38	c
39	b
40	a

41	c
42	c
43	c
44	d
45	b
46	b
47	c
48	c
49	c
50	c